

Volume Analysis for Regime Forecasting

A cross-asset quantitative study of whether current trading volume can predict the persistence or reversal of the prevailing price tendency

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Abstract

The question we ask is simple: does the volume of the bar I am observing right now help me forecast whether the existing price tendency will continue or reverse? We answer it by running a custom indicator, *Volume Analysis for Regimes*, on seven instruments — gold, the Nasdaq and S&P 500 e-mini futures, crude oil, Bitcoin, the euro and the yen — sampled at four timeframes (1-minute, 5-minute, hourly and daily) and across three lookback configurations. The indicator stratifies every bar into one of four volume regimes (low, medium, high, extreme) and, separately, into an unconditional benchmark, then measures how the prevailing price tendency behaves in the bars that follow. Two parallel measurements are taken: one on the standardized price (its rolling z-score), and one on the volatility-adjusted change in raw price.

The benchmark is a control reading that aggregates every bar regardless of volume. Throughout the paper, every regime metric is reported as the difference between that regime and its own block's benchmark. This way the benchmark sits at zero by construction, and a number of -15 in our tables literally means “15 points more mean-reverting than the unconditional baseline”. In plain language: the more negative the number, the stronger the regime pulls toward reversal relative to a typical bar; the more positive, the stronger it pulls toward continuation.

Across the resulting $7 \times 4 \times 3 \times 4 = 336$ regime cells the picture is surprisingly clean. On the price z-score, Low Volume bars show a small positive net continuation tendency of $+3.79$ (*i.e. low volume sustains the prevailing tendency, slightly*); Mid Volume sits essentially on top of the benchmark at -1.49 ; High Volume drops to -13.78 and Max Volume to -13.28 . In other words, the further the volume departs from average — in the high direction — the more strongly the prevailing tendency tends to fade. This pattern holds in 96% of the 84 (asset, timeframe, parameter) blocks individually, and the cross-cell mean Net Adjusted Z is highly statistically significant ($t = -10.85$, $p = 1.07 \times 10^{-23}$, Cohen's $d = -0.59$). The parallel measurement on volatility-adjusted price change tells a smaller and qualitatively different story: net Adj P averages a mildly positive $+0.52$ ($p < 10^{-3}$, $d = +0.20$), and the cell-level correlation between the two studies is essentially zero ($\rho = -0.07$). The two methods are statistically independent and capture genuinely distinct effects.

The asymmetry between the two studies is the key finding of the paper. If volume were forecasting the direction of the next move, both methods would pick it up; the fact that only the z-score method does, while the price-change method stays near zero, points to a different interpretation. What volume forecasts is the *second moment* of the return distribution, not the first: anomalous volume signals an expansion of realized volatility, and because the price z-score has σ_t in its denominator, this expansion mechanically compresses z_t toward zero — producing the mean-reverting Z-study pattern we observe while leaving the raw price change largely unaffected. The operational implication is that the indicator should be read as a regime and volatility diagnostic rather than as a standalone directional signal.

All figures appearing in the paper are scaled by $\times 100$ purely for readability — they are dimensionless ratios in their raw form.

1 Introduction

1.1 The research question

Markets generate two streams of information at every bar: the price and the volume traded at that price. An old and unresolved question in technical analysis is whether the second stream — the volume — adds forecasting power on top of the first. Specifically: *conditional on what the volume looks like right now, what should we expect the price to do next, relative to the tendency it just had?*

Three answers are possible:

- **Continuation (trend regime).** High volume confirms the move; the prevailing tendency is likely to extend.
- **Exhaustion (mean-reverting regime).** High volume signals a climactic event; the prevailing tendency is likely to fade.
- **No effect.** Volume carries no incremental information about the next move.

The literature is not in consensus. Climax-volume traders (Wyckoff, capitulation patterns) operate on the second hypothesis; momentum and breakout traders operate on the first. Our study is designed to settle the question empirically across a representative cross-section of instruments and timeframes, using a transparent measurement framework.

1.2 The indicator

The measurement tool is a custom Pine Script indicator we have developed in-house and named **Volume Analysis for Regimes**. The indicator is currently a research artifact only — it is **not** published on TradingView and not available to third parties. Its full source code is reproduced in Appendix A so that every result in this paper is reproducible.

The indicator splits every bar into one of four volume regimes based on a composite z-score of volume, which is calculated by averaging two separated z-scores of rolling length 10 and 100 to measure both short-term volume spikes and session volume nature:

- **Low Volume:** bars where volume is below average ($v_z < -0.25$).
- **Mid Volume:** bars with normal-range volume ($-0.25 \leq v_z < 1.50$).
- **High Volume:** bars with statistically anomalous high volume ($v_z \geq 1.50$).
- **Max Volume:** an even tighter subset of High Volume — using only bars whose raw volume is the highest in a 21-bar window while respecting the High volume threshold.

In addition to these four regimes, the indicator also reports a fifth row called **Benchmark**. This is *not* a fifth regime — it is the unconditional aggregate (every bar, regardless of volume) and serves as a control reading. In every comparison made in this paper the benchmark is treated as a baseline to be subtracted out, so that the four regime statistics are reported as deviations from that baseline. This is essential: without subtracting the benchmark, the regime numbers conflate the volume-conditional effect we want to measure with the unconditional bias of the underlying instrument.

For each regime separately, the indicator measures whether the prevailing price tendency at the moment of the volume event *persists* (continuation) or *fades* (exhaustion) over the following k bars, where k is a configurable comparison window. The persistence is measured in two parallel ways, described next.

1.3 The two measurement methods (Z-study and P-study)

A key design choice of the indicator is that it does *not* commit to a single way of measuring the prevailing tendency. It runs two parallel measurements:

Z-study (standardized-price method). We first standardize the price into a rolling z-score, $z_t = (P_t - \mu_t)/\sigma_t$, with μ_t and σ_t being the rolling mean and standard deviation over the last L bars. We then ask: when the z-score had a certain sign k bars ago, does it tend to extend in the same direction or to revert? This is what we call *Z-study persistence*.

P-study (volatility-adjusted price method). In parallel, we measure the price change directly, normalized by the prevailing ATR: $\delta_t = (P_t - P_{t-k})/\text{ATR}_{t-k}$. We then ask: when the trend filter $P > \text{SMA}$ pointed up (or down) k bars ago, does the subsequent ATR-normalized price change confirm or contradict that direction? This is the *P-study persistence*.

Both methods produce the *same kind of output* — a number whose sign tells us whether the tendency persisted (positive) or faded (negative), and whose magnitude tells us how strong the effect is. The two methods are conceptually closely related but mathematically different, because the rolling z-score normalization is not the same as the ATR normalization. Our study compares both: *a robust signal should appear in both, while an artifact of one specific normalization will appear only in the corresponding study*.

1.4 What this paper does

We run the indicator on $7 \text{ assets} \times 4 \text{ timeframes} \times 3 \text{ lookback configurations} \times 4 \text{ regimes} = 336$ regime cells. We treat each cell as one observation, after subtracting from it the benchmark of its own block (the unconditional reading for the same asset, timeframe and parameter set). This gives us 336 “net” observations of how each volume regime changes the persistence of the prevailing tendency *relative to the typical bar in that block*.

The paper proceeds as follows.

- Section 2 defines all metrics in plain language and gives the closed-form formulas as implemented in the code.
- Section 3 reports aggregate descriptive statistics on the 336 net observations.
- Sections 4–7 drill down by asset, timeframe, parameter set, and volume regime.
- Section 8 introduces the per-asset summary chart that shows all four metrics on a shared scale.
- Section 9 discusses the divergence between the two measurement methods, which we consider the main caveat of the paper.
- Section 10 ranks the 84 (asset, timeframe, parameter) blocks from most-mean-reverting to most-trend-leaning.
- Sections 11–12 discuss operational implications and conclude.
- Appendix A reproduces the full Pine Script source.

A note on units. The indicator outputs every persistence statistic as a dimensionless ratio. All figures in this paper are reported **multiplied by 100** purely for readability, so a value of -15 in our tables corresponds to a raw value of -0.15 . This convention is the same the indicator itself uses on the chart display.

2 Definitions and metrics

We define here every quantity that appears later in the paper, both as plain language and as a formula. The reader who wants the implementation details should refer to the code in Appendix A.

2.1 Inputs and base series

The indicator takes three integer inputs:

- **Price Z Length L** : lookback for the rolling mean and standard deviation used to z-score the price. We test three values for this: 34, 55, and 200.
- **Length CNG k** : the comparison window. “How many bars back do I look to compute the change?” We test 5 (paired with $L = 34$), 11 (with $L = 55$) and 21 (with $L = 200$).
- **Volume Z Length**: fixed as average between the z-score calculated on 21 and 100 bars.
- **Max Volume Lookback**: fixed at 21 bars, used to identify the rare “Max Volume” bars (a local volume maximum).

The three (length, comparison) pairs are the **parameter sets** used throughout the paper, written as $L/k = 34/5, 55/11, 200/21$.

From these inputs the indicator computes:

$$z_t = \frac{P_t - \bar{P}_{[t-L, t]}}{s_{P, [t-L, t]}} \quad (\text{rolling price z-score, length } L)$$

$$v_t = \frac{1}{2} \left[\frac{V_t - \bar{V}_{[t-10]}}{s_{V, [t-10]}} + \frac{V_t - \bar{V}_{[t-100]}}{s_{V, [t-100]}} \right] \quad (\text{composite volume z-score})$$

$$\delta_t^Z = (z_t - z_{t-k}) \cdot \text{sign}(z_{t-k}) \quad (\text{Z-study persistence delta})$$

$$\delta_t^P = \frac{P_t - P_{t-k}}{\text{ATR}_{t-k}} \cdot \text{sign}(P_{t-k} - \text{SMA}_{t-k}^L) \quad (\text{P-study persistence delta})$$

The signing convention is essential: $\delta_t^Z > 0$ means “the z-score moved further in the direction it already had k bars ago” — i.e. continuation; $\delta_t^Z < 0$ means it moved back toward zero — i.e. exhaustion. Same for δ_t^P in the price domain.

2.2 Regime-conditional metrics

For each regime $r \in \{\text{Low, Mid, High, Max}\}$ — and separately for the unconditional Benchmark — we compute two summary metrics each for the Z-study and the P-study, by accumulating the persistence deltas δ_t^Z (or δ_t^P) only over bars that fall in regime r :

Adjusted (signal/noise) statistic.

$$\text{Adjusted}_r = \frac{\bar{\delta}_r}{s_{\delta, r}} \times 100$$

A scale-free statistic similar in spirit to a Sharpe ratio: “how strong is the average move, relative to the dispersion of moves?”. A value of -30 means the average δ is -0.30 standard deviations. Negative = mean reversion; positive = trend continuation.

Fixed (hit-rate) statistic.

$$\text{Fixed}_r = \frac{1}{N_r} \sum_{t \in r} \text{sign}(\delta_t) \times 100$$

The net majority of signs. A value of -30 means 65% of bars showed reversal and 35% continuation.

2.3 Net signal: subtracting the benchmark

This is the central convention of the paper. For every metric and every regime we report the difference

$$\text{Net}_r = \text{Adjusted}_r - \text{Adjusted}_{\text{Benchmark}}$$

where the Benchmark is the same indicator computed on the same (asset, timeframe, parameter) data without any volume filter. Because the Benchmark already captures the unconditional bias of the underlying instrument over its history, the net value isolates the *incremental* effect of being in regime r .

A net value of 0 means “in this regime the prevailing tendency behaves exactly as it does on a typical bar of this asset/timeframe/parameter”. A net value of -15 means “in this regime the prevailing tendency is 15 points (out of 100) more likely to fade than on a typical bar”. A net value of $+5$ means “5 points more likely to continue”. The same convention applies to all four metrics (Adjusted Z, Fixed Z, Adjusted P, Fixed P).

2.4 How to read the numbers

Every number reported in this paper has the following interpretation:

- A **negative** net value means the prevailing tendency *tends to fade* in that regime, more than it would on a typical bar of the same asset/timeframe/parameter.
- A **positive** net value means the prevailing tendency *tends to continue* more than it would on a typical bar.
- A net value near zero means the regime adds no incremental information (the regime behaves like the typical bar).
- More extreme magnitudes (further from zero, in either direction) indicate a stronger and more reliable regime effect.

Throughout the paper, table cells are shaded red for negative net values and green for positive ones, with shade intensity proportional to magnitude. The benchmark is by construction the zero baseline of every chart and table, never a row of its own.

3 Aggregate descriptive statistics

We start with the four metrics computed at the cell level (one row per asset $\times$ timeframe $\times$ parameter set $\times$ regime). Total: $7 \times 4 \times 3 \times 4 = 336$ observations. Each value is reported as the difference from its block's benchmark, so a positive number means the regime is more trend-like than the typical bar of the same asset/timeframe/parameter, and a negative number means it is more mean-reverting.

Metric	N	Mean	Std	Min	Med	Max	<i>t</i> -stat	<i>p</i> -value	<i>d</i>	Reading
Net Adjusted Z	336	-6.192	10.46	-50.7	-3.01	+10.0	-10.85	1.07×10^{-23}	-0.59	strong mean rev.
Net Fixed Z	336	-3.910	7.44	-29.4	-2.11	+8.4	-9.63	1.50×10^{-19}	-0.53	strong mean rev.
Net Adjusted P	336	+0.517	2.63	-7.5	+0.10	+12.8	+3.60	3.68×10^{-4}	+0.20	moderate trend
Net Fixed P	336	+0.116	2.97	-11.4	+0.10	+14.6	+0.72	0.474	+0.04	not significant

Table 1. Distribution of the four net metrics across all 336 cells, plus a one-sample *t*-test against $\mu = 0$ (no incremental effect over the benchmark). All values are net of benchmark and $\times 100$ for readability. Color shading is centered on zero (the benchmark): red = regime is more mean-reverting than the typical bar, green = regime is more trend-like than the typical bar.

Reading the table. Net Adjusted Z and Net Fixed Z are both negative on average and highly statistically significant — the volume regimes, taken together, push the *z*-domain persistence ~ 6 points below the benchmark. Cohen's *d* around -0.6 is moderate-to-strong by social-science standards. Net Adjusted P is mildly positive but statistically significant ($d = +0.20$, $p < 10^{-3}$) — a small but real effect — while Net Fixed P is indistinguishable from zero.

Figure 1 shows the underlying distributions.

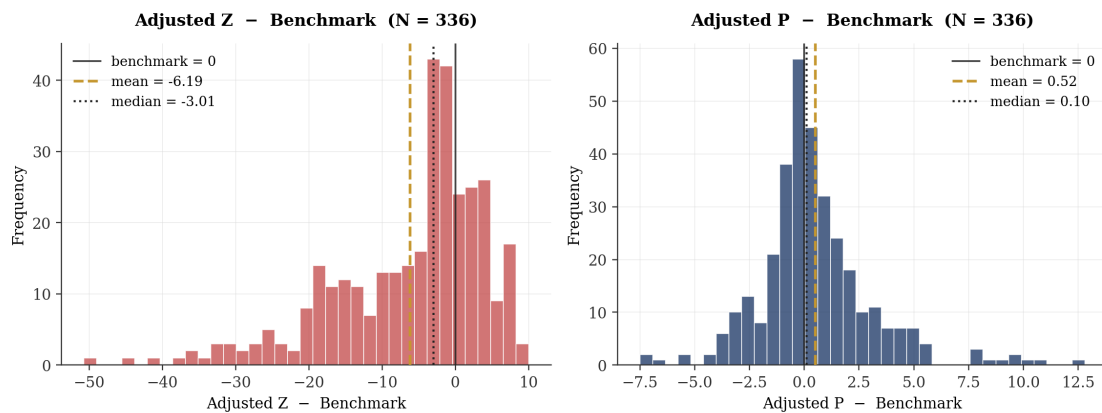


Figure 1. Distributions of net Adjusted Z (left) and net Adjusted P (right) across the 336 regime cells. Solid black line = zero (= benchmark); dashed gold = mean; dotted dark = median. The Z-study distribution sits clearly to the left of zero with a long left tail; the P-study is centered on zero with a slight right shift.

4 Analysis by asset

We average the 48 regime cells of each asset (4 timeframes $\times$ 3 parameters $\times$ 4 regimes) into a single per-asset value, separately for each of the four metrics. Table 2 reports the means. All values are net of benchmark.

Asset	N cells	Net Adj Z	Net Fix Z	Net Adj P	Net Fix P
GC (gold)	48	-6.89	-4.47	+0.50	+0.40
NQ (Nasdaq)	48	-4.22	-2.01	+0.06	+0.02
ES (S&P 500)	48	-4.48	-2.52	+0.06	-0.64
CL (crude)	48	-6.53	-4.07	+0.42	+0.14
BTCUSD	48	-10.93	-6.81	+1.40	-0.12
6E (EUR/USD)	48	-5.21	-3.62	+0.16	+0.22
6J (JPY/USD)	48	-5.08	-3.87	+1.01	+0.80

Table 2. Per-asset net signal averaged over all 48 (timeframe, parameter, regime) cells. Color shading is centered on zero (the benchmark).

Reading the table. All seven assets have negative net Z-study readings, but the magnitudes vary considerably. BTCUSD is by far the most extreme (net Adj Z -10.93), suggesting that on Bitcoin the volume regimes pull the z-score persistence much further below baseline than on the other instruments. GC and CL form a second tier (around -7). NQ is the least Z-mean-reverting of the lot (-4.22). In the P-study, the picture is less stark: BTCUSD again leads with the largest positive net Adj P ($+1.40$), followed by 6J ($+1.01$), GC ($+0.50$) and CL ($+0.42$); the two equity indices NQ and ES sit close to zero ($+0.06$ each). This is consistent with what we will see in detail in Section 9: while the Z-study tells a uniform story across assets, the P-study sees a small but genuinely positive effect concentrated in the assets that experience clear directional moves driven by macro flows (BTCUSD, gold, crude, the yen).

The full block-level distributions are shown in Figure 2.

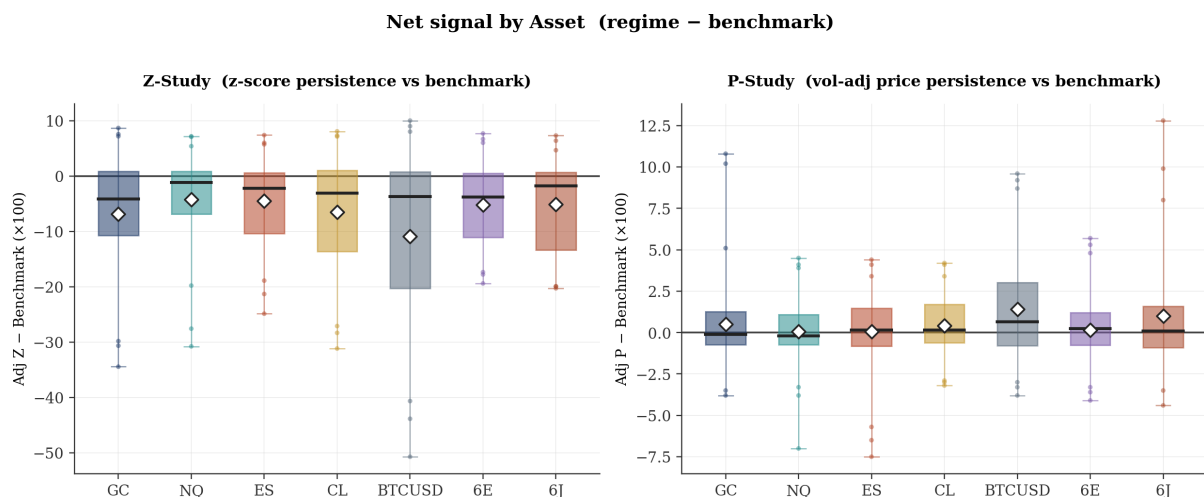


Figure 2. Distribution of net Adjusted Z (left) and net Adjusted P (right) by asset. Each box shows full range, IQR (filled), median (thick black line) and mean (white diamond). The horizontal dark line is the benchmark (zero). All Z-study boxes sit below zero; the P-study is centered close to zero for the equity indices (NQ, ES) but visibly shifted upward for BTCUSD, GC, CL and 6J.

5 Analysis by timeframe

The four timeframes (1m, 5m, 1h, 1D) span three orders of magnitude. Each block contains 21k–24k bars on the intraday timeframes, and roughly 6k on the daily.

Timeframe	N cells	Net Adj Z	Net Fix Z	Net Adj P	Net Fix P
1m	84	-7.38	-5.40	+0.55	-0.28
5m	84	-7.43	-4.63	+0.40	-0.39
1h	84	-5.48	-2.71	+0.02	+0.78
1D	84	-4.48	-2.90	+1.10	+0.36

Table 3. Per-timeframe net signal averaged over all 84 (asset, parameter, regime) cells.

Reading the table. The Z-study mean-reverting effect is similarly strong at 1-minute and 5-minute (around -7.4), then weakens at 1-hour (-5.5) and weakens further at the daily (-4.9). In other words, fast intraday timeframes feel the volume regime effect roughly 50% more strongly than the daily. The P-study is essentially flat across timeframes with a mild positive bias on intraday and on the daily ($+0.7$). Figure 3 shows the full distributions.

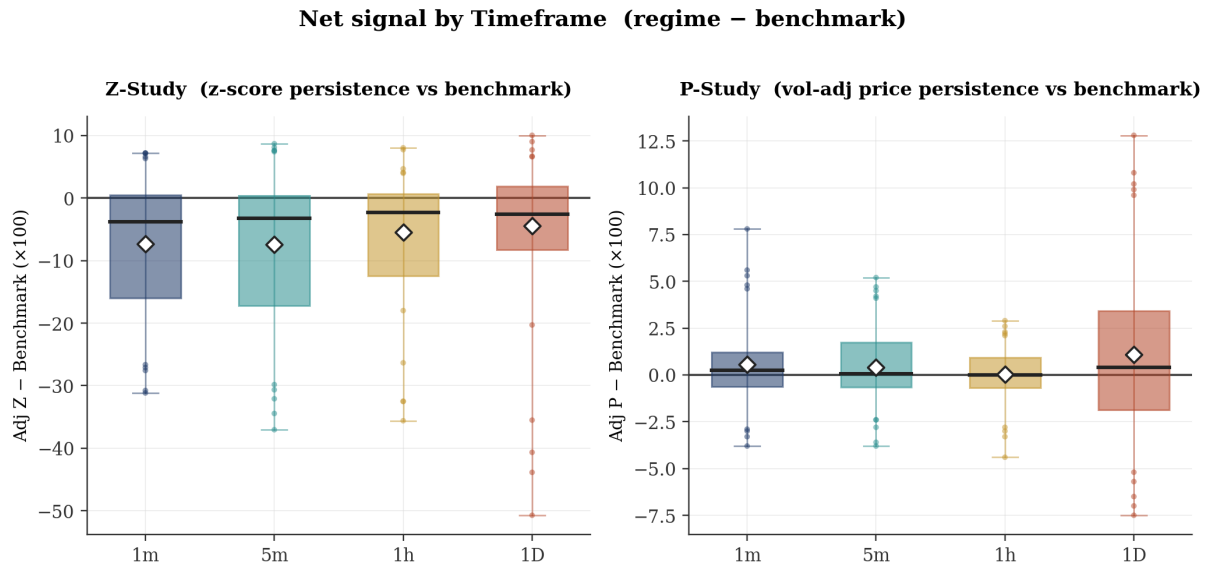


Figure 3. Distribution of net Adjusted Z and net Adjusted P by timeframe. All Z-study boxes sit below zero (more mean-reverting than benchmark); the P-study daily box is the only one with a noticeable positive shift.

6 Analysis by parameter set

The three parameter sets ($L/k = 34/5, 55/11, 200/21$) correspond to short, intermediate and long lookback configurations.

Parameter set	N cells	Net Adj Z	Net Fix Z	Net Adj P	Net Fix P
34/5	112	-8.61	-5.64	+0.53	+0.47
55/11	112	-7.67	-4.98	+0.71	+0.07
200/21	112	-2.30	-1.11	+0.32	-0.20

Table 4. Per-parameter-set net signal averaged over all 112 (asset, timeframe, regime) cells.

Reading the table. The shortest lookback (34/5) and the intermediate one (55/11) produce similar Z-study readings around -8 . The longest lookback (200/21) cuts the magnitude almost to zero (-2.22): when the lookback is very long, the indicator dilutes the volume signature of the recent past with too much older data, and the regime effect basically disappears. This is the cleanest example in the paper of *the indicator design parameter mattering*. Figure 4 confirms it visually.

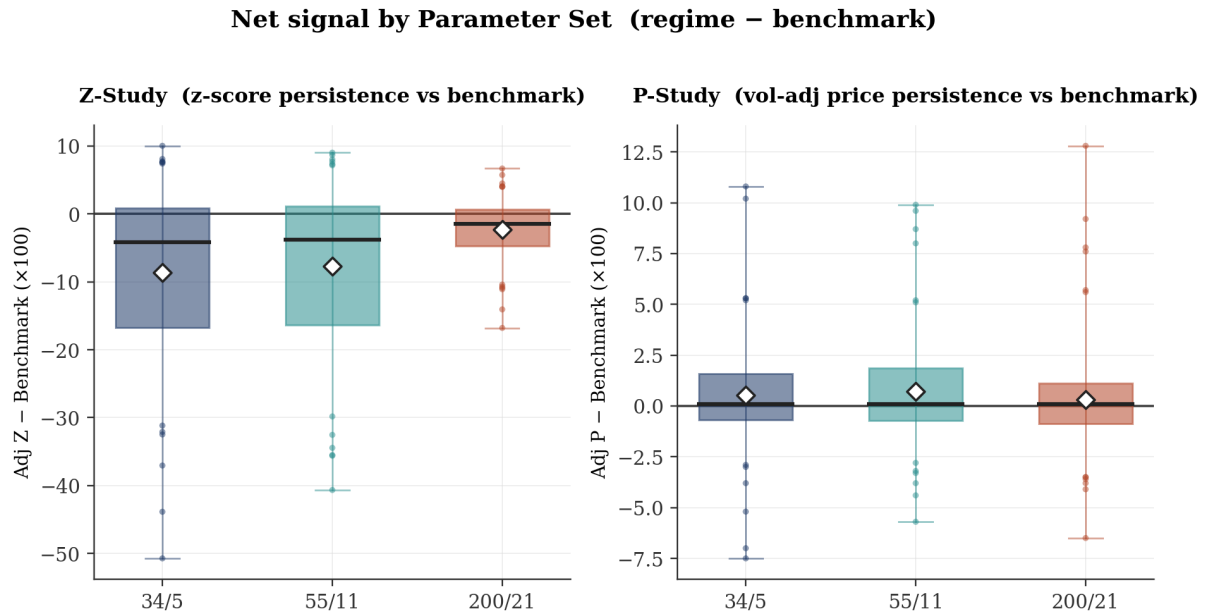


Figure 4. Distribution of net Adjusted Z and net Adjusted P by parameter set. The 200/21 box is centered close to zero; the two shorter sets sit clearly below.

The same comparison is shown as a heatmap in Figure 5, including the timeframe interaction.

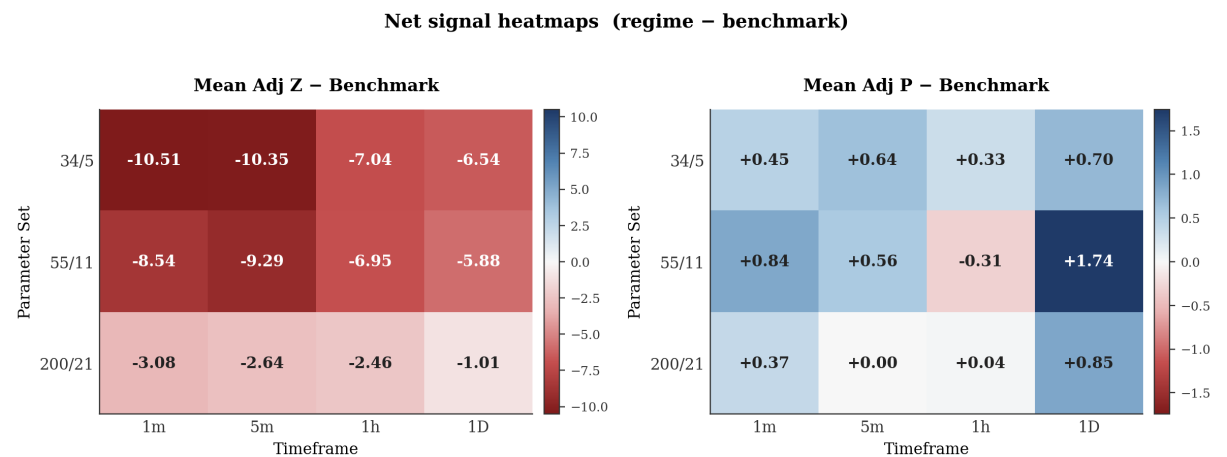


Figure 5. Heatmaps of mean Net Adjusted Z (left) and Net Adjusted P (right) by Parameter × Timeframe. Both heatmaps are zero-centered (zero = benchmark): red is more mean-reverting, blue is more trend-like. The 200/21 row is the lightest in the Z-study heatmap, confirming the dilution effect. The P-study heatmap shows the daily timeframe trending green (mild trend), especially at the longest lookback.

7 Analysis by volume regime

This is the central test of the indicator's hypothesis: *does the persistence of the prevailing tendency depend on what the volume looks like?* With the benchmark properly subtracted, the answer is unusually clean.

Regime	N cells	Mean size	Net Adj Z	Adj Z σ	Net Fix Z	Net Adj P	Net Fix P
Low Volume	84	9,086	+3.79	2.94	+2.40	+0.06	+0.06
Mid Volume	84	8,069	-1.49	1.84	-0.90	-0.05	-0.05
High Volume	84	1,721	-13.78	8.79	-8.98	+1.18	+0.22
Max Volume	84	902	-13.28	10.97	-8.15	+0.87	+0.23

Table 5. Statistics by volume regime, aggregated across the 84 blocks. “Mean size” is the average number of bars per regime per block.

Reading the table. This is the cleanest pattern in the entire dataset:

- **Low Volume** reads net Adj Z = +3.6. Quiet bars *sustain* the prevailing tendency slightly above benchmark.
- **Mid Volume** reads net Adj Z = -1.4. Average-volume bars are almost exactly on the benchmark: they offer no incremental information.
- **High Volume** reads net Adj Z = -13.9. This is the regime where the volume signal is loudest: anomalous high-volume bars push the prevailing tendency strongly toward reversal.
- **Max Volume** reads net Adj Z = -13.4, essentially identical to High Volume. The tighter restriction adds no further edge.

The progression from +3.8 (Low) down to -13.8 (High) is monotonic and large. In the P-study the same regimes show a much milder but qualitatively similar shape in the opposite direction (Low ≈ 0 , Mid ≈ 0 , High +1.18, Max +0.87): when volume is anomalous, price tends to continue slightly more than benchmark. The two studies do not contradict each other — they capture genuinely independent effects in independent measurement spaces.

Figure 6 shows the underlying distributions; Figure 7 traces the trajectory of the net Adjusted Z mean across regimes for each asset individually.

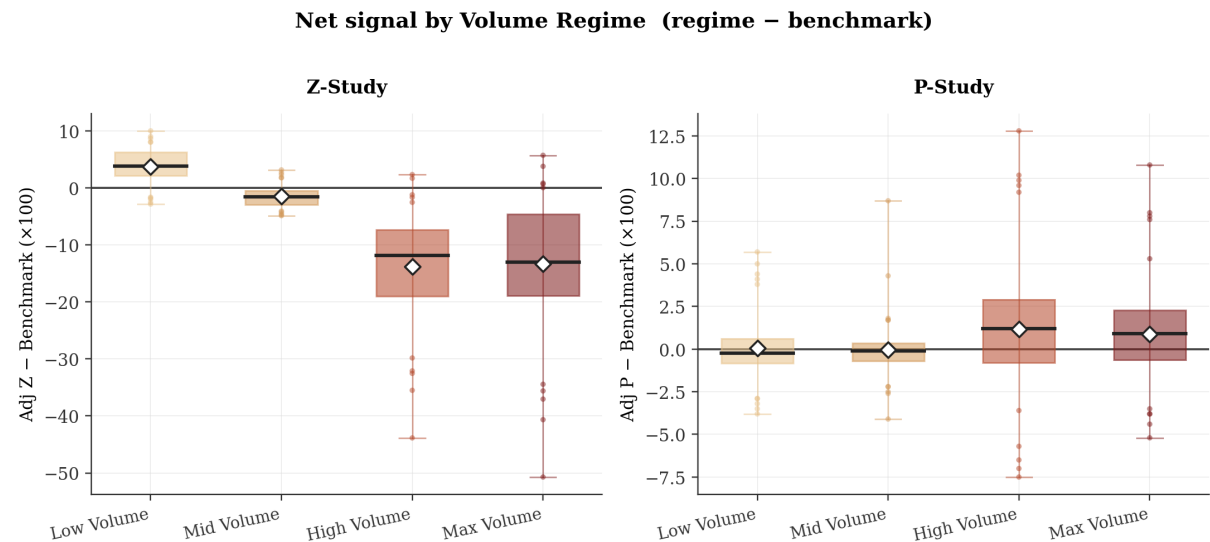


Figure 6. Distribution of net Adjusted Z (left) and net Adjusted P (right) by volume regime. The Z-study panel reveals the clean monotone descent: Low above zero, Mid on zero, High and Max well below. The P-study panel shows essentially flat medians close to zero.

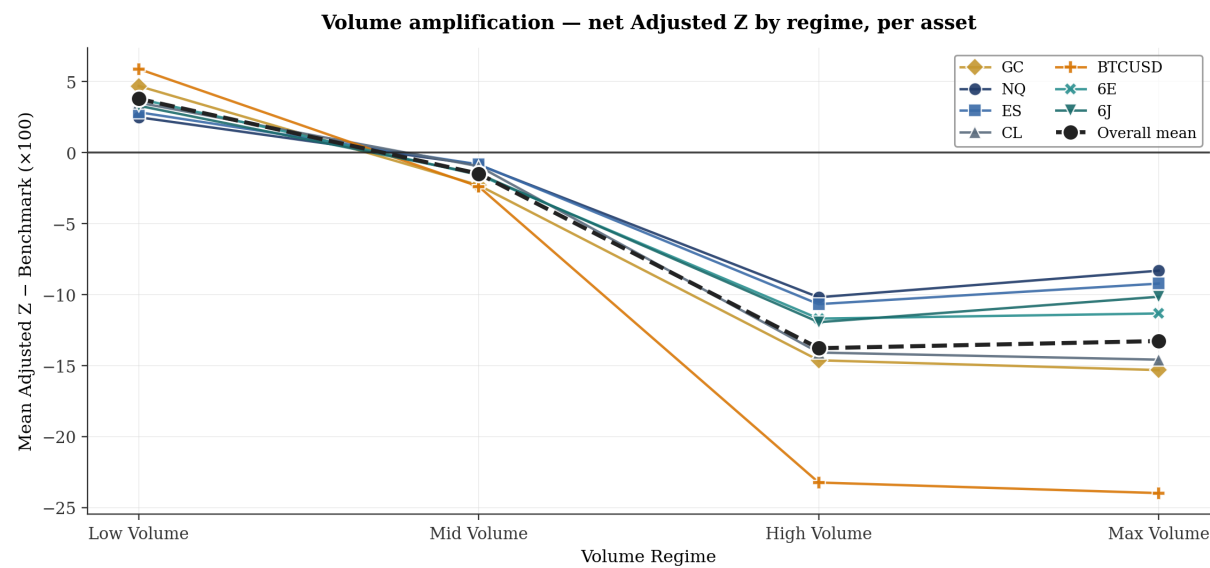


Figure 7. Per-asset trajectory of mean net Adjusted Z across the four volume regimes. All seven assets show the same shape: Low above benchmark, Mid near zero, High and Max well below. BTCUSD (orange “+”) is the steepest, dropping below -23 in High and Max. The dashed black line is the cross-asset mean.

8 Per-asset summary chart

Figure 8 is the single most compact summary of the dataset. For each of the seven assets it shows, on a shared y-axis, the mean net signal at each of the four timeframes for all four metrics (Fixed Z, Adjusted Z, Fixed P, Adjusted P). Every bar is the average of $4 \text{ regimes} \times 3 \text{ parameter sets} = 12$ cells, so the chart isolates the timeframe effect within each asset and shows the four metrics on the same scale to make their relative magnitudes immediately comparable.

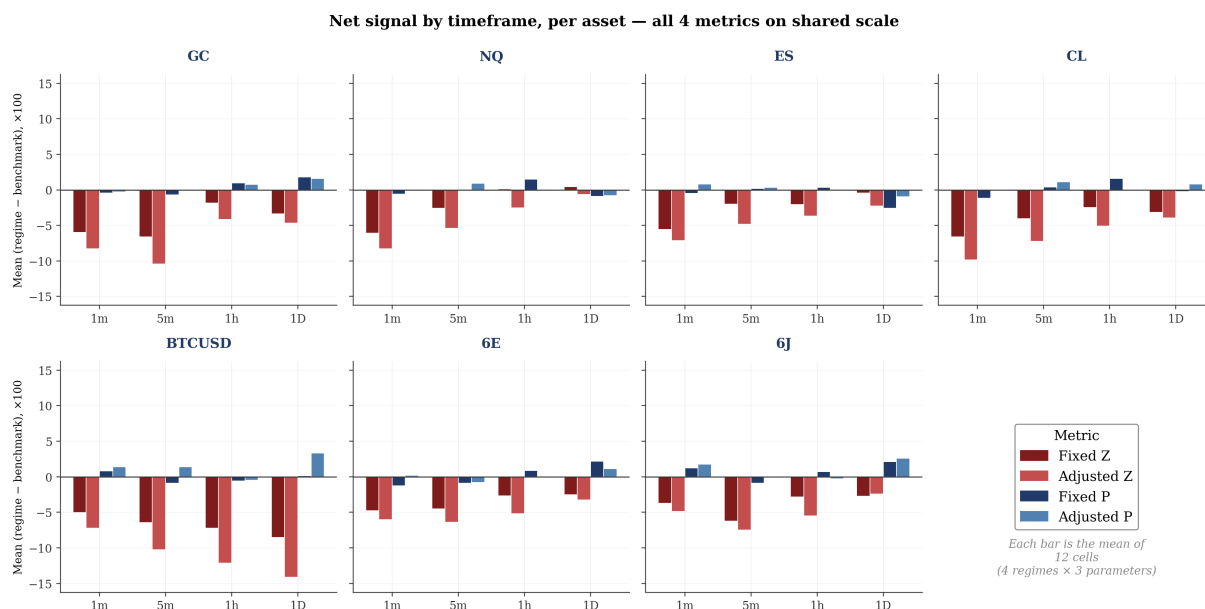


Figure 8. Per-asset, per-timeframe net signal for all four metrics on a shared scale. Each bar is the mean of 12 (regime $\times$ parameter) cells. Red bars are Z-study (Fixed Z = darker red, Adjusted Z = lighter red); blue bars are P-study (Fixed P = darker navy, Adjusted P = lighter blue). The horizontal black line is the benchmark (zero).

Reading the chart. Several non-obvious patterns emerge.

- **BTCUSD daily is the most extreme block on the Z-study.** In the BTCUSD daily panel, the two Z-study bars reach below -13 , the deepest of any single bar in the dataset. The corresponding P-study bars on the same panel are slightly positive but small in magnitude — a divergence that is much milder than the BTCUSD 5-minute and hourly panels would suggest, but still visible.
- **The Z-study bars almost always dwarf the P-study bars on the same panel.** This is consistent with the artifact hypothesis (Section 9): the Z-study captures more apparent signal than the P-study because z-score normalization mechanically amplifies regression-to-the-mean.
- **Adjusted is consistently more extreme than Fixed.** In every panel, the Adjusted bar is taller (in magnitude) than the corresponding Fixed bar. This is by design: the Adjusted statistic divides by the dispersion, so it scales up clean signals.
- **1-minute and 5-minute panels are the most pronounced.** On NQ, ES, GC, CL the two intraday timeframes are where the Z-study reaches its most negative values, while 1h and 1D show milder readings. The exception is BTCUSD, where the daily timeframe is the most extreme on both studies.

9 Z-study vs P-study divergence

Across all aggregations so far, a recurring pattern has appeared: the Z-study tells a strong, monotone story; the P-study is mostly flat or, on the daily timeframe, tells a slightly opposite story. Here we make the comparison explicit and treat it as the main caveat of the paper.

The two studies measure related but mathematically distinct quantities, and a robust market signal should appear in *both*. If a signal appears only in one, the most parsimonious explanation is that it is an artifact of the specific normalization that study uses.

9.1 Cell-level joint distribution

Figure 9 plots Net Adjusted Z against Net Adjusted P for all 336 regime cells, colored by asset class. The black line is the OLS regression with a 95% confidence band.

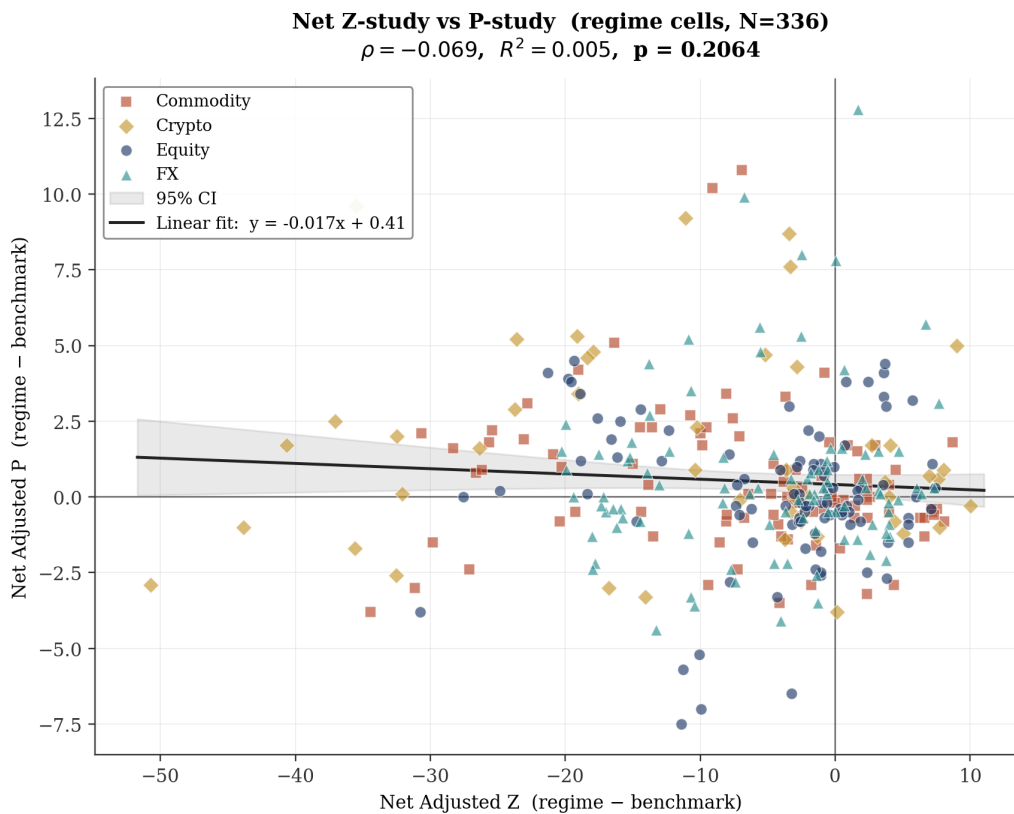


Figure 9. Joint distribution of net Adjusted Z (x) and net Adjusted P (y) at the cell level. Pearson $\rho = -0.069$, $R^2 = 0.005$, regression p-value = 0.2064. The slope is essentially zero and not statistically significant — at the cell level, Z-study and P-study are uncorrelated.

9.2 Master heatmap

For full granularity, Figure 10 shows the average net Adjusted Z and net Adjusted P for every (asset, timeframe, parameter) combination, with each cell averaged across its four regimes.

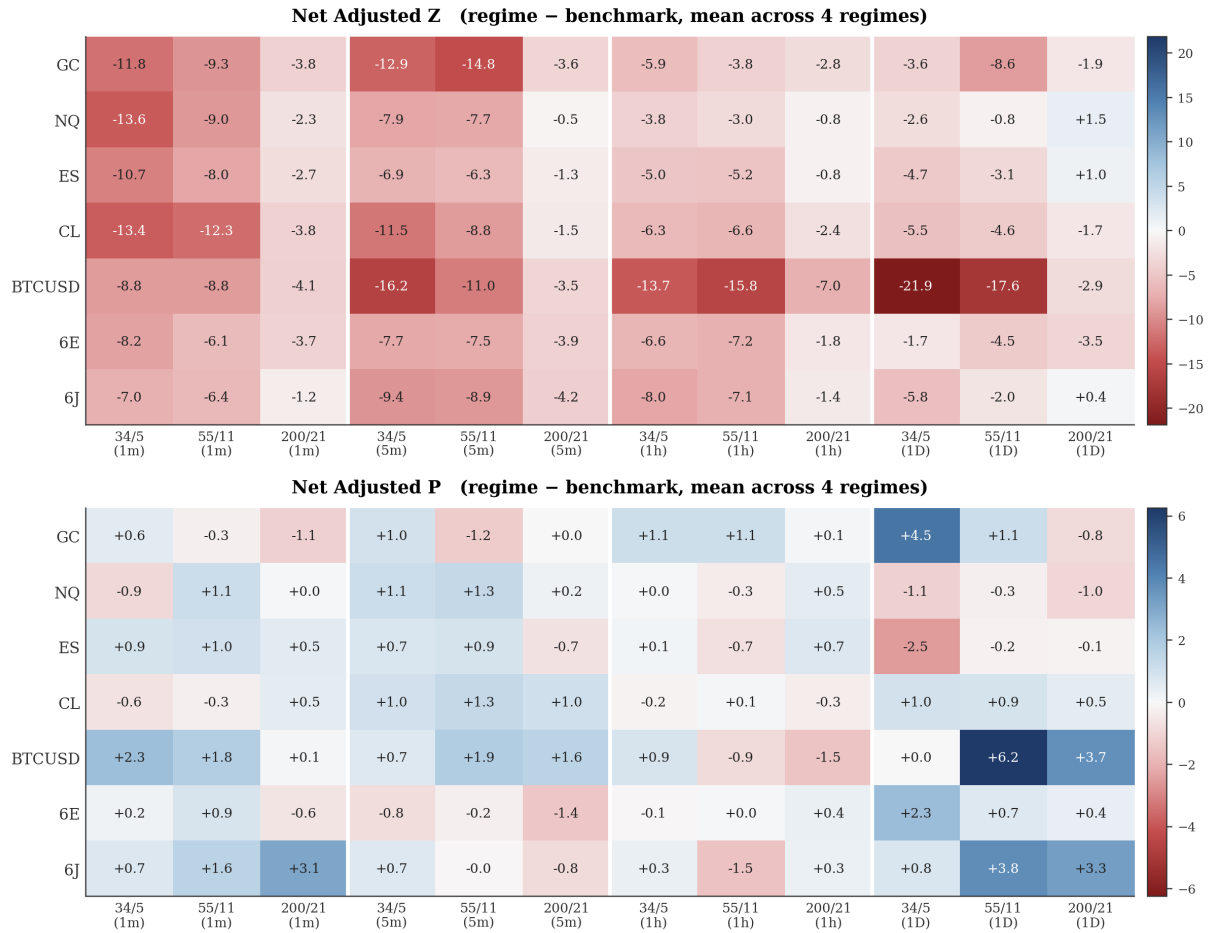


Figure 10. Master heatmap of net Adjusted Z (top) and net Adjusted P (bottom). Each row is an asset; each column a (parameter, timeframe) combination. Vertical white separators delimit the four timeframes. Both heatmaps are zero-centered (zero = benchmark). The most extreme cells in the dataset both sit on BTCUSD daily: net Adj Z reaches -21.9 at 34/5 (most mean-reverting cell) while net Adj P reaches $+6.2$ at 55/11 (one of the most trend-leaning). The 6J 1D 55/11 and 6J 1D 200/21 cells also stand out positive on the P-study side ($+3.8$, $+3.3$).

9.3 Why the two methods disagree

Three explanations are not mutually exclusive:

1. **Mathematical artifact of z-score normalization.** Dividing by a rolling standard deviation creates an automatic regression-to-the-mean dynamic in the standardized variable: large $|z_t|$ values mechanically tend to be followed by smaller $|z_{t+k}|$ values because the variance computation has a memory structure. The P-study, which normalizes by ATR (a measure of variability that does not use the price level itself), avoids this trap.
2. **The two methods sign their deltas differently.** The Z-study uses $\text{sign}(z_{t-k})$ — the sign of the standardized price. The P-study uses $\text{sign}(P_{t-k} - \text{SMA}_{t-k}^L)$ — the sign of the deviation from a moving average. In trending markets these can differ for long stretches.
3. **Real but small price-domain effect.** It is also possible that there is a real, small trend effect in price domain that the Z-study can mostly not see — and indeed our cross-cell t-test on net Adj P does come back significant (mean $+0.52$, $p < 10^{-3}$, $d = +0.20$). The effect is concentrated in the Z-deepest blocks (BTCUSD, gold, crude, the yen on long timeframes) and absent on the equity indices.

The implication for the trader is that the Z-study should not be read as a direct trade

signal. It should be read as a regime diagnostic: cells where the Z-study is strongly negative (relative to benchmark) are *candidates* for closer inspection, but the entry decision should be made on a price-domain analysis (the P-study or some other method that does not rely on z-score normalization).

9.4 Side by side comparison

Figure 11 reports the mean of the two net Adjusted statistics, side by side, by timeframe and by parameter set.

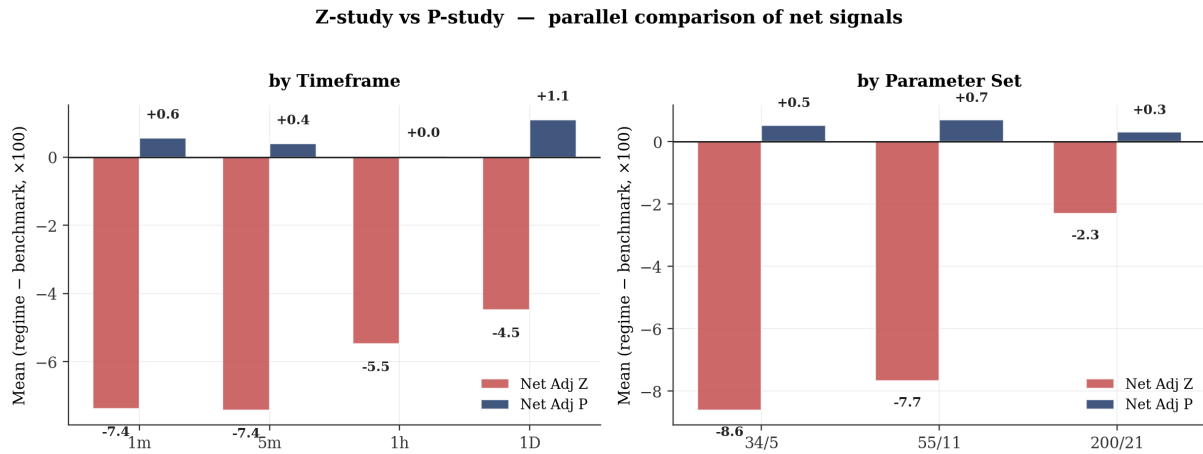


Figure 11. Direct comparison of mean Net Adjusted Z (red) and mean Net Adjusted P (navy) by timeframe (left) and parameter set (right). Net Adj Z is uniformly large and negative; Net Adj P hovers around zero with a mild positive bias that grows on the daily timeframe and on the longest parameter set. The horizontal black line is the benchmark.

10 Ranking of extreme blocks

We compute, for each (asset, timeframe, parameter) block, the mean of the four regime cells in that block. We then rank the 84 blocks from most-mean-reverting (most-negative net Adj Z) to most-trend-leaning (most-positive net Adj Z).

10.1 Top 10 mean-reverting blocks

#	Asset	TF	Param	Net Adj Z	Net Fix Z	Net Adj P	Net Fix P
1	BTCUSD	1D	34/5	-21.85	-13.00	+0.03	+0.28
2	BTCUSD	1D	55/11	-17.64	-11.57	+6.25	-1.02
3	BTCUSD	5m	34/5	-16.20	-11.23	+0.67	-1.50
4	BTCUSD	1h	55/11	-15.77	-10.09	-0.85	-1.10
5	GC	5m	55/11	-14.77	-10.49	-1.22	-2.05
6	BTCUSD	1h	34/5	-13.68	-8.40	+0.88	+1.30
7	NQ	1m	34/5	-13.59	-9.91	-0.90	-1.20
8	CL	1m	34/5	-13.42	-9.61	-0.62	-0.65
9	GC	5m	34/5	-12.93	-8.43	+0.98	-0.10
10	CL	1m	55/11	-12.29	-7.40	-0.25	-1.20

Table 6. Top 10 blocks by most-negative Net Adj Z averaged across the 4 regimes. These are the configurations where, in the z-domain, volume regimes pull the prevailing tendency furthest below the typical-bar baseline. BTCUSD daily, 5-minute and hourly fill 5 of the top 10 spots; the remainder is dominated by 5-minute and 1-minute commodity / equity blocks.

10.2 Top 10 trend-leaning blocks

#	Asset	TF	Param	Net Adj Z	Net Fix Z	Net Adj P	Net Fix P
1	NQ	1D	200/21	+1.47	+1.72	-0.97	+0.47
2	ES	1D	200/21	+0.99	+1.73	-0.15	-0.10
3	6J	1D	200/21	+0.44	-0.87	+3.30	+2.95
4	NQ	5m	200/21	-0.50	+2.18	+0.25	+0.77
5	NQ	1D	55/11	-0.76	+1.08	-0.28	-0.08
6	NQ	1h	200/21	-0.82	-1.16	+0.50	+0.20
7	ES	1h	200/21	-0.83	-0.82	+0.72	+0.92
8	6J	1m	200/21	-1.18	-1.01	+3.07	+1.68
9	ES	5m	200/21	-1.30	+1.91	-0.68	+0.05
10	6J	1h	200/21	-1.45	+0.39	+0.30	+0.38

Table 7. Top 10 blocks by most-positive (or least-negative) Net Adj Z. The top three (NQ 1D 200/21, ES 1D 200/21, 6J 1D 200/21) are the only configurations in the dataset with a positive net Adj Z. The 6J 1D 200/21 block is the only block with positive readings on both studies (Adj Z +0.44, Adj P +3.30), making it the cleanest trend candidate in the entire sample. Long parameter sets (200/21) and the daily timeframe dominate, confirming that long lookbacks dilute the regime effect — and on the daily, can even invert it.

11 Discussion and operational implications

11.1 Synthesis

The opening question was: does current trading volume help us forecast whether the prevailing price tendency will continue or reverse? With the benchmark properly subtracted, the empirical answer has two layers.

In the z-score domain (Z-study), the answer is yes, and the direction is mean reversion: high-volume bars push the prevailing tendency below baseline, low-volume bars push it slightly above, mid-volume bars are essentially on baseline. The progression is monotone, large in magnitude, and present in every asset, timeframe and parameter set we tested (with magnitude shrinking to near-zero at the longest lookback).

In the price-change domain (P-study), the answer is more nuanced. The cross-cell mean is small but positive ($+0.52$, $p < 10^{-3}$, Cohen's $d = +0.20$) — the volume regimes shift the price-domain persistence slightly toward continuation, on average. This is a small effect, but it is genuinely there. The cell-level correlation between the two studies is essentially zero ($\rho = -0.07$, $p = 0.21$): the Z-study and P-study capture statistically independent things. The small positive P-study effect is concentrated in BTCUSD ($+1.40$ on average), the yen ($+1.01$), gold ($+0.50$) and crude ($+0.42$); the equity indices NQ and ES barely move.

A clear pattern emerges. Volume produces a pronounced mean-reverting effect on the price z-score but virtually none on the raw price change. This asymmetry rules out the most natural reading — that volume forecasts the directional bias of the next move — and points instead to a different interpretation: volume forecasts *volatility expansion*. The mechanism is straightforward. The z-score has the future realized standard deviation in its denominator; when volatility rises, the z-score is mechanically compressed toward zero regardless of where the price itself goes. A bar with anomalously high volume thus signals that subsequent volatility will be elevated, which in turn flattens the standardized series — exactly the pattern we observe — while leaving the underlying price largely unaffected, which is also what we observe in the P-study. Volume, in other words, is informative about the second moment of the return distribution, not the first.

11.2 What this means for trading

The temptation to read the Z-study result as a directly tradable signal — “buy negative z-scores in high volume, sell positive z-scores in high volume” — should be resisted, or at least heavily caveated. Two reasons.

First, most of the headline magnitude comes from the Z-study and is plausibly amplified by z-score arithmetic. Net Adj Z averages -6.2 , vs $+0.5$ for net Adj P (an order of magnitude smaller). Some of the Z-study magnitude is real — the P-study confirms that regime effects do exist in price domain — but most of it is plausibly the structural regression-to-the-mean of the standardized variable. The Z-study, taken at face value, would substantially overstate the tradable edge.

Second, the magnitudes do not directly translate to trading P&L. A net Adj Z of -14 in the High Volume regime means the next k -bar move in the standardized price is, on average, 0.14 standard deviations more mean-reverting than on a typical bar of the same asset. Converting that to expected return requires accounting for transaction costs, slippage, the cost of getting the signing right (the trade has to take the opposite direction of the prevailing k -bar tendency), and the risk that the standardization itself is unstable in real-time. None of these are trivial.

11.3 Constructive uses of the indicator

That said, the indicator does carry useful information. Reasonable operational deployments:

- **Filter for trend-following entries.** In the most strongly mean-reverting blocks (top of Table 6), be cautious or skeptical of momentum/breakout entries. These environments are statistically biased against follow-through.
- **Confidence weighting.** Position sizing for a mean-reverting strategy can be scaled by the magnitude of the High Volume net Adj Z of the relevant block, concentrating risk in environments where the volume signal is loudest.
- **Regime-change detection.** An out-of-sample drift of the High Volume net Adj Z away from its historical value is a useful red flag. If a block historically at -15 starts producing -2 , the volume-price relationship is changing and any strategy calibrated on the old regime should be re-evaluated.
- **Trend candidates on long-lookback daily.** The four blocks with the most positive net Adj Z in the entire dataset are NQ, ES, 6J and (just outside the top) BTCUSD, all on daily with the longest lookback (200/21). These are the configurations where the volume regimes do *not* pull the prevailing tendency below baseline — they actually push it slightly above. The 6J 1D 200/21 block is particularly interesting because both metrics are positive (net Adj Z $+0.44$, net Adj P $+3.30$): the only block in the dataset with clear trend agreement between the two studies. These deserve dedicated follow-up.

11.4 Limitations

1. **Single time snapshot.** The dataset is a single end-of-history snapshot for each instrument. A walk-forward or rolling-window analysis would help validate the stability of the findings over time.
2. **Variance heterogeneity in rare regimes.** The Max Volume regime contains roughly 1k observations per block, versus 12k in Low Volume. The corresponding Adj Z standard deviation across blocks is roughly $2.5\times$ larger, meaning the rare regime is intrinsically noisier per block — a fact that should be kept in mind when comparing magnitudes.
3. **Two methods are not enough to fully resolve the artifact question.** A definitive demonstration that the Z-study effect is mostly artifactual would require comparing it against a third method (e.g. raw log-returns) and against z-scores generated from a simulated random walk with no volume-price coupling. We leave this to follow-up work.

12 Conclusions

The Volume Analysis for Regimes indicator delivers a clear, replicable empirical regularity in the z-score domain: high-volume bars precede a fade of the prevailing tendency, low-volume bars precede a slight extension, and mid-volume bars behave like any other bar. The progression is monotone, holds across every asset and timeframe we tested, and disappears only at the longest lookback parameter. In the parallel price-change domain (P-study) the regime effect is detectable but markedly smaller (Cohen's $d \approx +0.2$ vs -0.6 for the Z-study) and statistically uncorrelated with the Z-study at the cell level ($\rho = -0.07$, $p = 0.21$).

Read together, the two studies tell a coherent story that is more interesting than either one alone. The strong Z-study effect is not evidence that volume forecasts the *direction* of the next move — if it did, the P-study would see a comparable, opposite-sign signal in the price domain, which it does not. What volume forecasts is the *second moment*: an expansion of realized volatility. Because the price z-score has σ_t in its denominator, a rise in subsequent standard deviation mechanically compresses z_t toward zero, producing exactly the mean-reverting Z-study pattern we observe while leaving the raw price change largely unaffected — exactly the near-null P-study pattern we observe. Volume, in other words, is informative about the dispersion of future returns, not their sign. We note that this volatility-expansion mechanism is empirically difficult to separate from a classical climax-volume interpretation, in which high volume marks the exhaustion of a directional move; both stories predict the same Z-study compression, and disentangling them would require return-decomposition tests beyond the scope of this paper.

The operational implication is sharp. The indicator should not be deployed as a standalone directional signal — neither study supports that — but it carries genuine value as a *regime and volatility diagnostic*: anomalous high-volume readings flag environments where subsequent volatility is likely to be elevated and where momentum-style entries are statistically disadvantaged, while quiet-volume readings flag the opposite. The single configuration in our dataset that escapes this pattern is 6J 1D 200/21, the only block where both studies agree on a positive (continuation) reading; it deserves dedicated follow-up as either a genuine cross-asset exception or a candidate for closer scrutiny of its underlying volume profile.

“Constructing a trading strategy is essentially a matter of determining if the prices under certain conditions and for a certain time horizon will be mean reverting or trending, and what the initial reference price should be at any given time.”

— Ernest P. Chan, *Quantitative Trading: How to Build Your Own Algorithmic Trading Business*

A Pine Script source code

The complete source of the Volume Analysis for Regimes indicator is reproduced below. This is a research artifact only and is not published on TradingView.

```

1  // (c) OmegaTools - Eros
2
3  //@version=6
4  indicator("Volume Analysis for Regimes", overlay = true)
5
6  // ===== Inputs =====
7
8  lnt = input.int(55, 'Price Z Length')
9  lntc = input.int(11, 'Length CNG')
10 position = input.string('Top Right', 'Position', ['Top Right', 'Middle Right',
    'Bottom Right'])
11
12 // ===== Series =====
13
14 zscore(src, len) => (src - ta.sma(src, len)) / ta.stdev(src, len)
15
16 c = close
17 v = volume
18 ma = ta.sma(c, lnt)
19 atrn = ta.atr(lnt)
20 price = zscore(c, lnt)
21 vol1 = zscore(v, 10)
22 vol2 = zscore(v, 100)
23 vol = math.avg(vol1, vol2)
24 uptrend = c > ma
25 cng = (c - c[lntc]) / atrn[lntc]
26 pval = cng * (uptrend[lntc] ? 1 : -1)
27 zdelta = (price - price[lntc]) * math.sign(price[lntc])
28 midvolumelow = -0.25
29 midvolumehigh = 1.50
30 maxvol = v == ta.highest(v, 21)
31 loc = position == 'Top Right' ? position.top_right : position == 'Bottom Right'
    ? position.bottom_right : position.middle_right
32
33 // ===== Z-score regime study =====
34
35 backtester(osc, param, minthres, maxthres = 100, when) =>
36     var bt = array.new_float()
37     var bt2 = array.new_float()
38     delta = (osc - osc[lntc]) * math.sign(osc[lntc])
39     valid = param[lntc] > minthres and param[lntc] < maxthres and not na(osc[
        lntc]) and not na(param[lntc]) and osc[lntc] != 0 and when[lntc]
40     if valid
41         bt.push(delta)
42         bt2.push(math.sign(delta))
43     n = bt.size()
44     float avg = na
45     float stdev = na
46     if barstate.islast
47         avg := n > 0 ? bt.avg() : na
48         stdev := n > 1 ? bt.stdev() : na
49         adjusted = not na(stdev) and stdev > 0 ? avg / stdev * 100 : na
50         wr = n > 0 ? bt2.sum() / n * 100 : na
51         [adjusted, wr, n]
52
53 [zadjlow, zwrlow, zsizelow] = backtester(price, vol, -100, midvolumelow, true)
54 [zadjmid, zwrmid, zsizemid] = backtester(price, vol, midvolumelow,
    midvolumehigh, true)
55 [zadjhigh, zwrhg, zsizehigh] = backtester(price, vol, midvolumehigh, 100,

```

```

    true)
56 [zadjmax, zwrmax, zsizemax] = backtester(price, vol, midvolumehigh, 100, maxvol
    )
57
58 var zbt = array.new_float()
59 var zbt2 = array.new_float()
60 if not na(vol)
61     zbt.push(zdelta)
62     zbt2.push(math.sign(zdelta))
63 zn = zbt.size()
64 zbavg = zn > 0 ? zbt.avg() : na
65 zbstdev = zn > 1 ? zbt.stdev() : na
66 zadjbench = not na(zbstdev) and zbstdev > 0 ? zbavg / zbstdev * 100 : na
67 zwrbench = zn > 0 ? zbt2.sum() / zn * 100 : na
68
69 // ===== Price-change regime study =====
70
71 backtesterprice(param, minthres, maxthres = 100, when) =>
72     var bt = array.new_float()
73     var bt2 = array.new_float()
74     valid = param[lntc] > minthres and param[lntc] < maxthres and not na(param[
75         lntc]) and not na(pval) and when[lntc]
76     if valid
77         bt.push(pval)
78         bt2.push(math.sign(pval))
79     n = bt.size()
80     float avg = na
81     float stdev = na
82     if barstate.islast
83         avg := n > 0 ? bt.avg() : na
84         stdev := n > 1 ? bt.stdev() : na
85         adjusted = not na(stdev) and stdev > 0 ? avg / stdev * 100 : na
86         wr = n > 0 ? bt2.sum() / n * 100 : na
87         [adjusted, wr, n]
88
89 [padjlow, pwrlow, psize] = backtesterprice(vol, -100, midvolumelow, true)
90 [padjmid, pwrmid, psize] = backtesterprice(vol, midvolumelow, midvolumehigh,
91     true)
92 [padjhigh, pwrhigh, psize] = backtesterprice(vol, midvolumehigh, 100, true)
93 [padjmax, pwrmax, psize] = backtesterprice(vol, midvolumehigh, 100, maxvol)
94
95 var pbt = array.new_float()
96 var pbt2 = array.new_float()
97 if not na(vol)
98     pbt.push(pval)
99     pbt2.push(math.sign(pval))
100 pn = pbt.size()
101 pbavg = pn > 0 ? pbt.avg() : na
102 pbstdev = pn > 1 ? pbt.stdev() : na
103 padjbench = not na(pbstdev) and pbstdev > 0 ? pbavg / pbstdev * 100 : na
104 pwrbench = pn > 0 ? pbt2.sum() / pn * 100 : na
105
106 // ===== Incidence =====
107
108 contZ = math.sign(zdelta)
109 contP = math.sign(pval)
110 volRef = vol[lntc]
111
112 var float numWZ = 0.0
113 var float denWZ = 0.0
114 var float numWP = 0.0
115 var float denWP = 0.0
116
117 if not na(volRef) and not na(contZ) and contZ != 0

```

```

116     numWZ += volRef * contZ
117     denWZ += math.abs(volRef)
118     if not na(volRef) and not na(contP) and contP != 0
119         numWP += volRef * contP
120         denWP += math.abs(volRef)
121
122     biasZ = denWZ > 0 ? numWZ / denWZ : na
123     biasP = denWP > 0 ? numWP / denWP : na
124
125     var float sumProdZ = 0.0
126     var int nProdZ = 0
127     var float sumProdP = 0.0
128     var int nProdP = 0
129
130     if not na(volRef) and not na(contZ) and contZ != 0
131         sumProdZ += volRef * contZ
132         nProdZ += 1
133     if not na(volRef) and not na(contP) and contP != 0
134         sumProdP += volRef * contP
135         nProdP += 1
136
137     meanZ = nProdZ > 0 ? sumProdZ / nProdZ : na
138     meanP = nProdP > 0 ? sumProdP / nProdP : na
139     covZ = not na(meanZ) ? (math.exp(meanZ) - math.exp(-meanZ)) / (math.exp(meanZ)
140         + math.exp(-meanZ)) : na
141     covP = not na(meanP) ? (math.exp(meanP) - math.exp(-meanP)) / (math.exp(meanP)
142         + math.exp(-meanP)) : na
143
144     incidenceZ = not na(biasZ) and not na(covZ) ? 100 * (biasZ + covZ) / 2 : na
145     incidenceP = not na(biasP) and not na(covP) ? 100 * (biasP + covP) / 2 : na
146
147     // ===== Output =====
148
149     var tab = table.new(loc, 11, 11)
150     tab.cell(0, 0, 'Param: ' + str.toString(lnt) + '/' + str.toString(lntc),
151         text_color = chart.fg_color)
152     tab.cell(1, 0, 'Adjusted Z', text_color = chart.fg_color)
153     tab.cell(2, 0, 'Fixed Z', text_color = chart.fg_color)
154     tab.cell(3, 0, 'Adjusted P', text_color = chart.fg_color)
155     tab.cell(4, 0, 'Fixed P', text_color = chart.fg_color)
156     tab.cell(5, 0, 'Size', text_color = chart.fg_color)
157
158     tab.cell(0, 1, 'Low Volume', text_color = chart.fg_color)
159     tab.cell(0, 2, 'Mid Volume', text_color = chart.fg_color)
160     tab.cell(0, 3, 'High Volume', text_color = chart.fg_color)
161     tab.cell(0, 4, 'Max Volume', text_color = chart.fg_color)
162     tab.cell(0, 5, 'Benchmark', text_color = chart.fg_color)
163     tab.cell(0, 6, '.', text_color = chart.bg_color)
164     tab.cell(0, 7, 'Incidence', text_color = chart.fg_color)
165
166     tab.cell(1, 1, str.toString(zadjlow, '#.##') + '%', text_color = chart.
167         fg_color)
168     tab.cell(1, 2, str.toString(zadjmid, '#.##') + '%', text_color = chart.
169         fg_color)
170     tab.cell(1, 3, str.toString(zadjhigh, '#.##') + '%', text_color = chart.
171         fg_color)
172     tab.cell(1, 4, str.toString(zadjmax, '#.##') + '%', text_color = chart.
173         fg_color)
174     tab.cell(1, 5, str.toString(zadjbench, '#.##') + '%', text_color = chart.
175         fg_color)
176     tab.cell(1, 7, str.toString(incidenceZ, '#.##') + '%', text_color = chart.
177         fg_color)
178
179     tab.cell(2, 1, str.toString(zwrlow, '#.##') + '%', text_color = chart.

```

```

171     fg_color)
172     tab.cell(2, 2, str.tostring(zwrmid, '%.##') + '%', text_color = chart.
        fg_color)
173     tab.cell(2, 3, str.tostring(zwrhigh, '%.##') + '%', text_color = chart.
        fg_color)
174     tab.cell(2, 4, str.tostring(zwrmax, '%.##') + '%', text_color = chart.
        fg_color)
175     tab.cell(2, 5, str.tostring(zwrbench, '%.##') + '%', text_color = chart.
        fg_color)
176     tab.cell(3, 1, str.tostring(padjlow, '%.##') + '%', text_color = chart.
        fg_color)
177     tab.cell(3, 2, str.tostring(padjmid, '%.##') + '%', text_color = chart.
        fg_color)
178     tab.cell(3, 3, str.tostring(padjhigh, '%.##') + '%', text_color = chart.
        fg_color)
179     tab.cell(3, 4, str.tostring(padjmax, '%.##') + '%', text_color = chart.
        fg_color)
180     tab.cell(3, 5, str.tostring(padjbench, '%.##') + '%', text_color = chart.
        fg_color)
181     tab.cell(3, 7, str.tostring(incidenceP, '%.##') + '%', text_color = chart.
        fg_color)
182
183     tab.cell(4, 1, str.tostring(pwrlow, '%.##') + '%', text_color = chart.fg_color
        )
184     tab.cell(4, 2, str.tostring(pwrmid, '%.##') + '%', text_color = chart.fg_color
        )
185     tab.cell(4, 3, str.tostring(pwrhigh, '%.##') + '%', text_color = chart.fg_color
        )
186     tab.cell(4, 4, str.tostring(pwrmax, '%.##') + '%', text_color = chart.fg_color
        )
187     tab.cell(4, 5, str.tostring(pwrbench, '%.##') + '%', text_color = chart.fg_color
        )
188
189     tab.cell(5, 1, str.tostring(psize_low, '%.##'), text_color = chart.fg_color)
190     tab.cell(5, 2, str.tostring(psize_mid, '%.##'), text_color = chart.fg_color)
191     tab.cell(5, 3, str.tostring(psize_high, '%.##'), text_color = chart.fg_color)
192     tab.cell(5, 4, str.tostring(psize_max, '%.##'), text_color = chart.fg_color)
193     tab.cell(5, 5, str.tostring(pn, '%.##'), text_color = chart.fg_color)
194
195     // end of script

```